4. UI/UX Design & Prototyping

The front end is a Power BI report with three pages: Triage, Patient and Executive. There is no web or mobile interface. The screens are designed around the data that exists in the five source tables, which sets three limits.

- **ed.triage** has no timestamp and no key to admissions. The Triage page therefore has no waiting-time column and orders records by risk tier and score.
- Triage vitals are one snapshot per record. Time-based charts on the Patient page use lab results (**hosp.labevents**), which carry *charttime*.
- ED registration time exists only in **hosp.admissions** (*edregtime*), and only for admissions that came through the emergency department. Arrival-pattern charts on the Executive page cover those admissions only.

Source table	Key	Content used by the report
ed.triage	stay_id	Vitals, pain, acuity (ESI), chief complaint. Triage and Patient pages.
hosp.patients	subject_id	Gender, anchor_age. Patient page.
hosp.admissions	hadm_id	Admission type, admit and discharge time, discharge location, death flag, ED registration and exit time. Patient and Executive pages.
hosp.labevents	labevent_id	Lab value, unit, reference range, abnormal flag, charttime. Patient page.
hosp.d_labitems	itemid	Lab name, fluid, category. Patient page.

1. Wireframes & Mockups

1. Report structure



Page tabs sit at the bottom-left of every page. Drill-through goes from a queue row to that patient's page.

Figure 4.1. Page flow of the report.

Page	User	Purpose	Main visuals	Fields
Triage	Triage nurse	Order records by risk and show which have no vitals.	Four tier KPI cards, priority queue table, records-by-tier chart, two slicers.	ed.triage: stay_id, subject_id, vitals, pain, acuity, chiefcomplaint. Calculated: risk_score, risk_tier.
Patient	Physician, analyst	Show one patient's profile, lab history and triage records.	Profile strip, two lab line charts, abnormal lab table, triage table, two slicers.	patients: gender, anchor_age. admissions: type, times, discharge_location. labevents: charttime, valuenum, valueuom, ref_range_lower/upper, flag. d_labitems: label.
Executive	Operations manager	Show load, length of stay and outcomes.	Four KPI cards, hourly registration chart, stay-length chart, admission-type chart, one slicer.	admissions: edregtime, admittime, dischtime, admission_type, hospital_expire_flag.

Risk tiers on the Triage page follow one rule set (draft bands).

Tier	Colour	Rule
High	Red	Score of 7 or more.
Medium	Amber	Score of 5 or 6, or any single vital scoring 3.
Low	Green	Score of 0 to 4 and no single vital scoring 3.
Unscored	Slate	Fewer than 3 of the 5 scored vitals present (heart rate, respiratory rate, oxygen saturation, systolic pressure, temperature).

4.1.2 Wireframes

Low-fidelity layouts. Dashed boxes are placeholders; numbers refer to the notes under each figure.

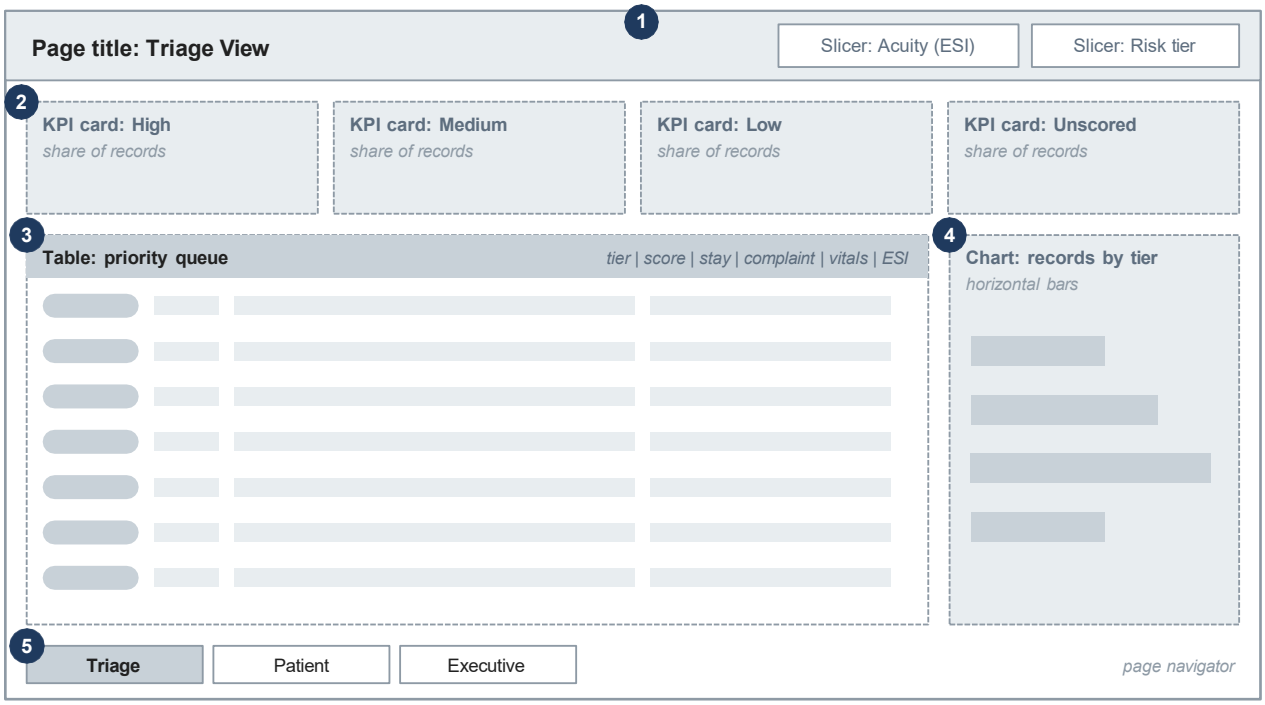


Figure 4.2. Wireframe, Triage page.

1. Slicers for ESI acuity and risk tier.
2. KPI cards: share of records in each tier.
3. Priority queue: tier, score, stay ID, chief complaint, vitals, ESI. Order: High, Unscored, Medium, Low; within a tier by score, highest first.
4. Bar chart of records by tier. Selecting a bar filters the queue.
5. Page navigator.

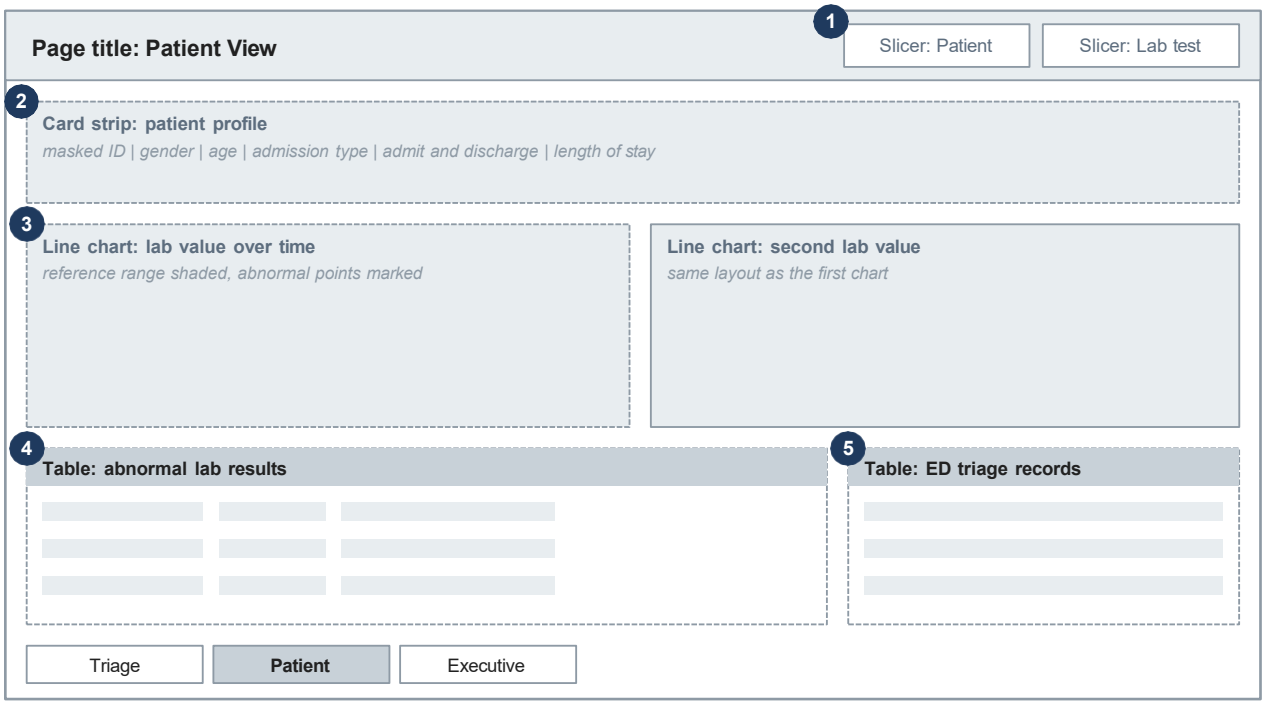


Figure 4.3. Wireframe, Patient page.

1. Slicers: patient (masked ID) and lab test.
2. Profile strip: gender, age, admission type, admit and discharge time, length of stay.
3. Two line charts of lab values over charttime. The shaded band is the reference range; abnormal results are marked.
4. Table of the latest abnormal result per lab test.
5. Table of the patient's ED triage records with their tier.

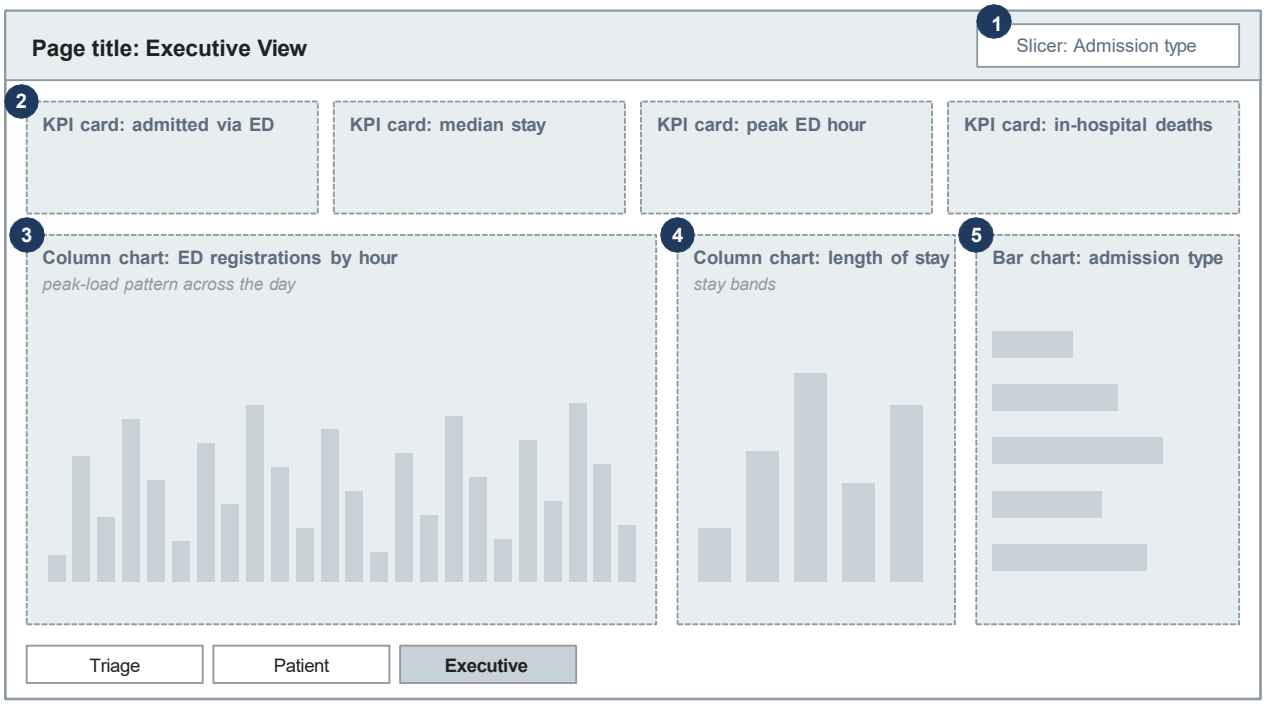


Figure 4.4. Wireframe, Executive page.

1. Slicer for admission type.

2. KPI cards: share of admissions that came through the ED, median stay, peak ED hour, share of in-hospital deaths.
3. Column chart of ED registrations by hour of day.
4. Column chart of length of stay in bands.
5. Bar chart of admission type.

4.2 UI/UX Guidelines

Design principles

- **Priority first.** The most urgent information sits top-left and is readable without scrolling.
- **One layout.** All pages share the same title bar, slicer position, card style and page navigator.
- **Detail on demand.** Summary on the Triage and Executive pages; detail by drill-through to the Patient page.
- **Missing data is shown.** A blank vital appears as a dash and the record is tiered Unscored, never Low.
- **Few visuals per page.** Five to six, so each can be read at a glance.
- **Colour carries meaning.** Tier colours are used for risk only.

Colour scheme



Colour	Use
Navy	Page title bar, table headers, active page tab.
Teal	Chart series, secondary KPI accents.
Canvas	Page background behind white cards.
Red, amber, green, slate	Risk tiers High, Medium, Low, Unscored. Not used elsewhere.
Text	Values and labels.

Typography

Segoe UI, the Power BI default, on a 1280 x 720 canvas (16:9). Figures in this document are scaled down.

Element	Size	Weight
Page title	18 pt	Semibold
Visual title	12 pt	Semibold
KPI value	26 pt	Bold
Table text and axis labels	10 pt minimum	Regular
Footnotes	9 pt	Regular, italic

Accessibility

- Every tier flag carries a text label as well as a colour.
- Alt text on every visual; a meaningful title on each.
- Tab order follows reading order: slicers, KPI cards, main visual, secondary visuals.
- Pages are checked in greyscale to confirm that tiers can still be told apart.

Data presentation rules

- Units on screen: temperature in degrees F (source unit; values below 50 are treated as Celsius and converted), heart rate in bpm, respiratory rate in breaths per minute, oxygen saturation in %, blood pressure in mmHg.

5. System Deployment & Integration

1. Technology Stack

Part	Tool	What it does
Frontend	Power BI Desktop	Shows the three screens.
Backend	Python 3 with pandas, NumPy, scikit-learn, Seaborn, Matplotlib	Reads and cleans the data, calculates the risk score, trains the classification model and draws exploratory charts.
Backend	Python logging, pytest	Keeps a record of every change to the data and runs the tests.
Database	SQL Server (local), database Mimic	Stores the raw tables, in one section (schema) per source folder, and the processed results in a section called dw.
Source	Five CSV files	The data supplied from MIMIC-IV.

The data comes from five files:

File	Contains
ed / triage	Vital signs, urgency level and main complaint at triage.
hosp / patients	Gender and age.
hosp / admissions	Admission type, admit and discharge times, where the patient went afterwards, whether they died in hospital.
hosp / labevents	Lab results with units, normal ranges and an abnormal flag.
hosp / d_labitems	Names of the lab tests.

Patients are the link between them. One patient can have several admissions, triage records and lab results. Lab results point to the lab test names, and most also point to an admission. Triage records have no date or admission number, so they cannot be tied to a particular admission.

Database design

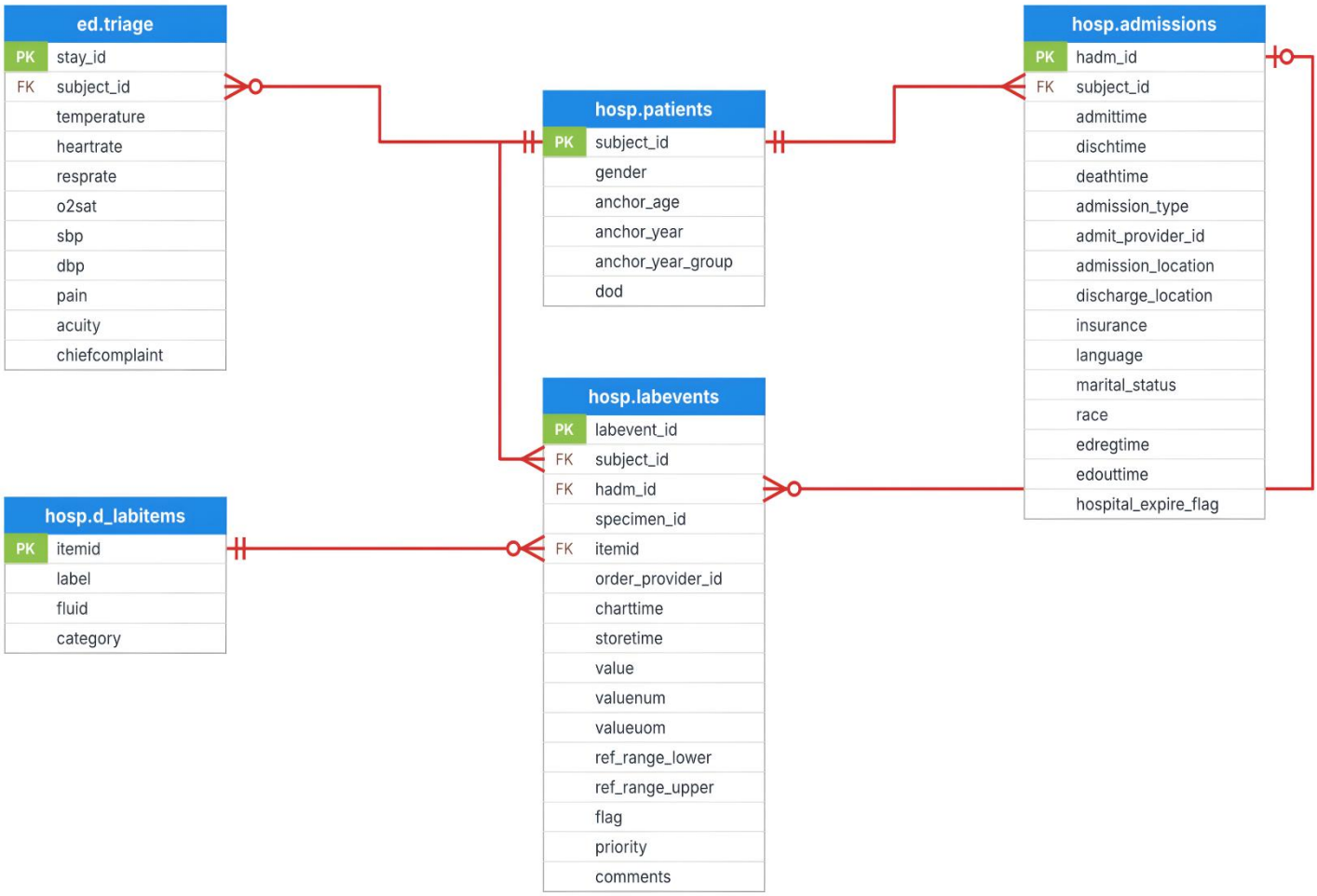


Figure 5.1. Entity-relationship diagram of the source tables.

Parent	Child	Key	Cardinality	Evidence in the data
hosp.patients	ed.triage	subject_id	1:M (Optional)	Not every patient has triage records; a patient can have several.
hosp.patients	hosp.admissions	subject_id	1:M (Mandatory)	Every patient has at least one admission and can have several.
hosp.patients	hosp.labevents	subject_id	1:M (Optional)	Every patient has lab results.
hosp.admissions	hosp.labevents	hadm_id	0..1 : M	hadm_id is empty on part of the lab rows, so the key is optional. Every non-empty value exists in admissions.
hosp.d_labitems	hosp.labevents	itemid	1:M (Optional)	Only part of the dictionary is used. Every itemid in labevents exists in d_labitems.
ed.triage	hosp.admissions	none	none	No shared key and no timestamp in triage. Related only through subject_id.

Primary keys (subject_id, hadm_id, stay_id, labevent_id, itemid) are unique in the files. The raw tables are loaded as text, so the database does not enforce keys on them; the integrity checks in the test plan do.

Processed tables in dw will use typed columns and primary keys.

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5.2 Deployment Diagram

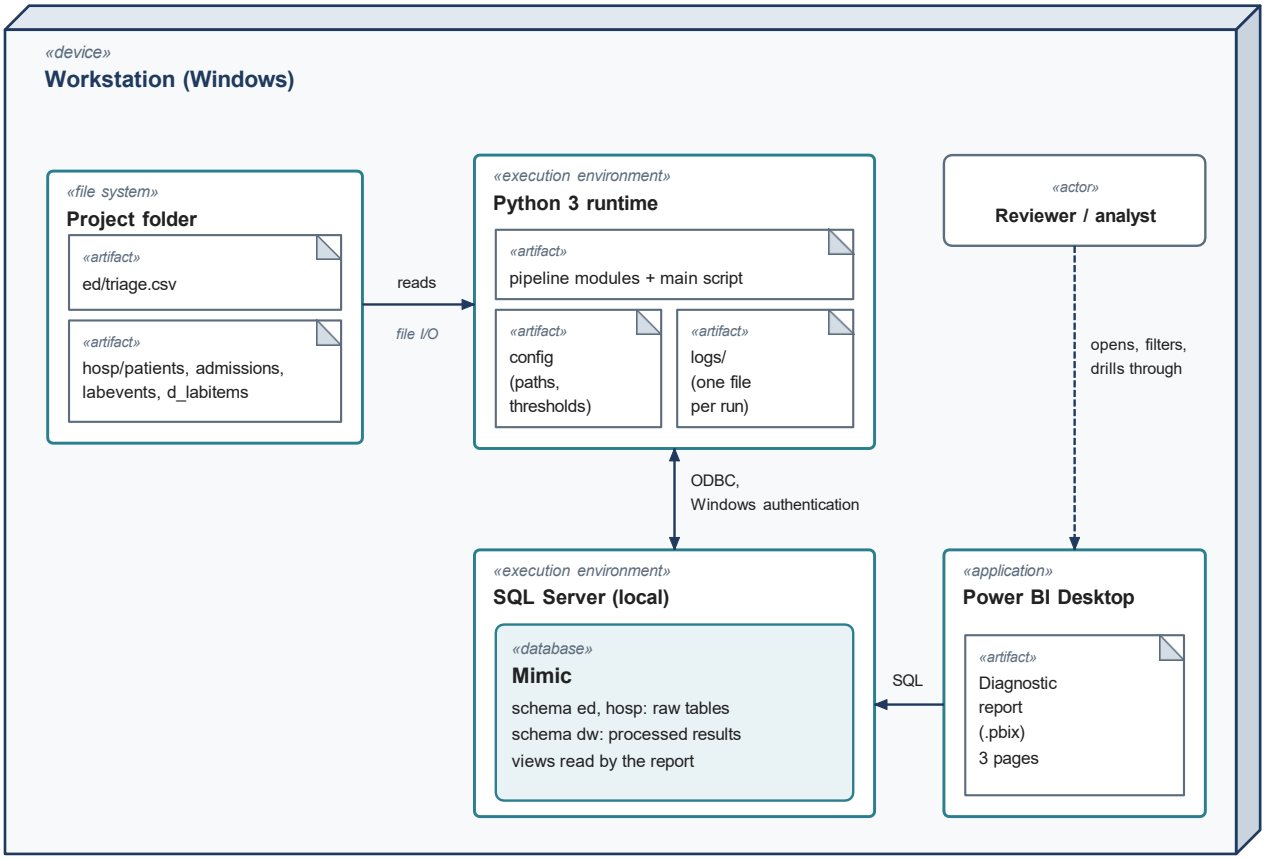


Figure 5.3. Deployment diagram (UML). All nodes run on one Windows workstation.

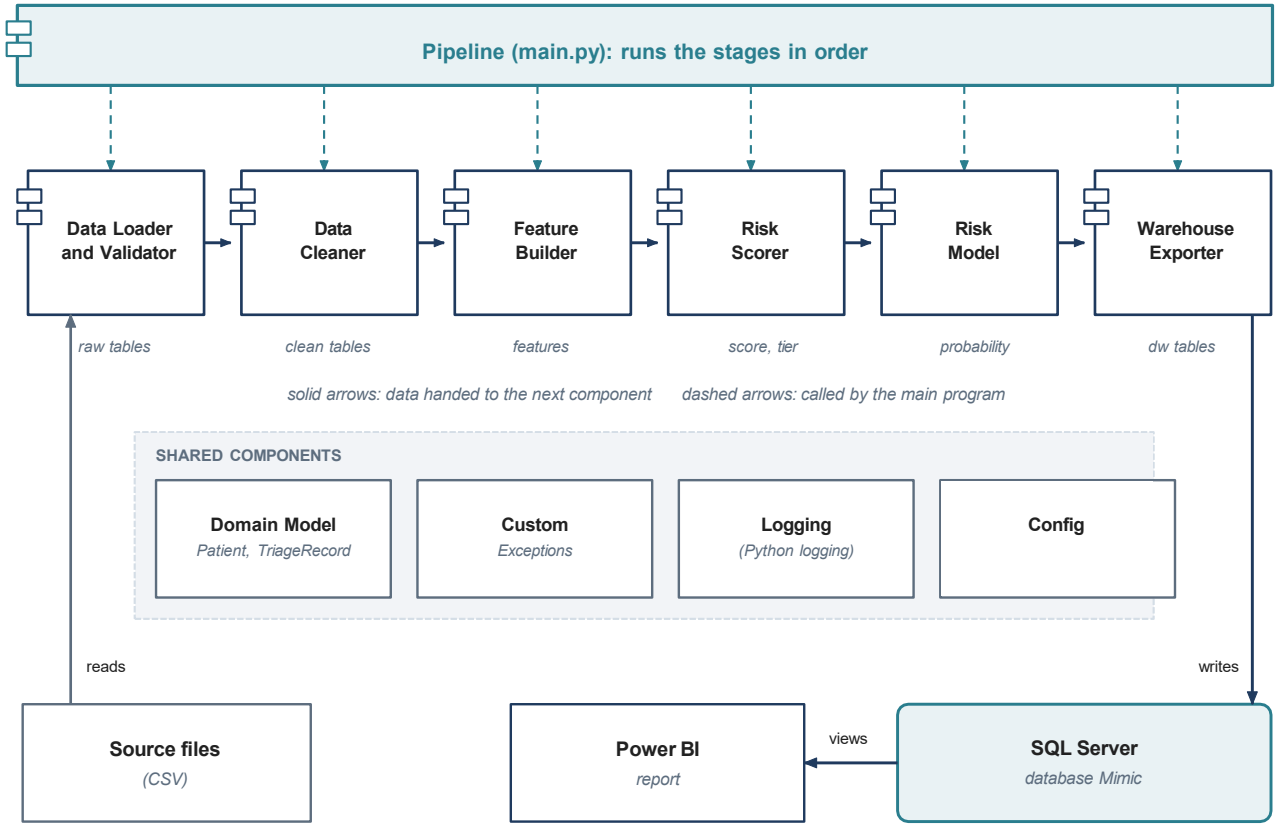
Element	Type	Description
Workstation (Windows)	Device	Single machine hosting every component. No network service is exposed.
Project folder	File system	ed/triage.csv and hosp/patients, admissions, labevents, d_labitems.
Python 3 runtime	Execution environment	Runs the pipeline modules and main script. Holds config (paths, server name, salt, thresholds) and writes one log file per run.
SQL Server (local)	Execution environment	Hosts database Mimic: raw schemas ed and hosp, result schema dw, and the views read by the report.

Element	Type	Description
Power BI Desktop	Application	Opens the .pbix report and queries the views.

Communication paths.

- Python reads the CSV files directly (file I/O).
- Python and SQL Server communicate over ODBC with Windows authentication, so no password is stored in the configuration.
- Power BI Desktop queries SQL Server through the SQL Server connector. Import or DirectQuery mode is still to be chosen.
- The report is refreshed by re-running the pipeline and then refreshing the data in Power BI.

5.3 Component Diagram



Interfaces between components are tables (pandas DataFrames in memory, SQL tables and views on disk).

Figure 5.4. Component diagram (UML). Solid arrows are data passed between components.

Component	Responsibility	Depends on	Hands over
Pipeline (main.py)	Calls each stage in order and stops on structural errors.	All components	Run summary
Data Loader and Validator	Reads the five CSV files and checks that expected columns exist.	Config, Exceptions, Logging	Raw tables
Data Cleaner	Applies plausibility and standardisation rules; logs every change.	Domain Model, Exceptions, Logging	Clean tables

6. Additional Deliverables

1. API Documentation

- The system has no API: no web service and no endpoints.
- Power BI reads three database views. **vw_triage_queue** feeds the Triage page (stay_id, masked patient key, chief complaint, vitals, pain, acuity, risk score, risk tier). **vw_patient_timeline** feeds the Patient page (patient, admission, lab results, triage records). **vw_exec_kpis** feeds the Executive page (ED registration time and hour, admission type, length of stay, discharge location, in-hospital death).
- Pipeline steps hand tables to each other: five CSV files, raw tables, clean tables, scored table, dw tables.

6.2 Testing & Validation

- **Unit tests:** the patient ID cannot be read directly; an impossible value (such as a diastolic pressure of 879) becomes missing and is logged; a temperature below 50 is converted from Celsius; invalid pain values become missing; known vitals give the expected score and tier; records with no vitals are Unscored.
- **Integration tests:** rows in equal rows out; triage to patients, labevents to admissions and labevents to d_labitems keys resolve; running the export twice gives the same counts; a full run from an empty dw schema finishes without manual steps.
- **Validation:** tier shares add up to 100% and report figures match SQL; High and Medium are more common at acuity 1 and 2 than at 3 and 4; a hand check of randomly chosen records.
- **User acceptance:** reviewers find the first record in the queue, filter by acuity, open a patient, find abnormal labs, find the busiest ED hour and read an Unscored record. A task passes when it is done without help.

6.3 Deployment Strategy

- **Environment:** one Windows workstation with local SQL Server, Python 3 and Power BI Desktop. Windows authentication.
- **Steps:** prepare the database (schemas ed, hosp, dw), load the CSV files into ed and hosp, run the pipeline, run the checks, refresh the report.
- **Scaling (not built):** indexes on subject_id, hadm_id and itemid and batch loading for larger extracts; Power BI service with scheduled refresh through a gateway for more users; a scheduled task to run the pipeline.